

The 8051 Microcontroller

Module 2

- I) Special function Registers(SFR's), Ports & memory organization of 8051
- II) Timer & UART of 8051 (8051 instruction set or embedded C to explain the concept)
- III) Under Actuator- DC motor Interfacing with 8051
- IV) under Sensor- Ultrasonic Interfacing with 8051
- V) A/D- Dual slope & D/A- 10 bit DAC

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- 8051 Microcontroller–Kenneth J. Ayala, Penram International/ Thomson, 3rd Edition, 2005.
 - The 8051 Microcontroller And Embedded Systems Using Assembly And C – Mazidi, Pearson Education India, 2nd edition, 2008. Jane W. S Liu, “ Real Time Systems” Pearson Higher Education ,3rd Edition, 2000.

Three criteria in Choosing a Microcontroller

- **meeting the computing needs of the task efficiently and cost effectively**

- speed, the amount of ROM and RAM, the number of I/O ports and timers, size, packaging, power consumption
- easy to upgrade
- cost per unit
- Noise of environment
- Power consumption

- **availability of software development tools**

- assemblers, debuggers, C compilers, emulator, simulator, technical support

- **wide availability and reliable sources of the microcontrollers**

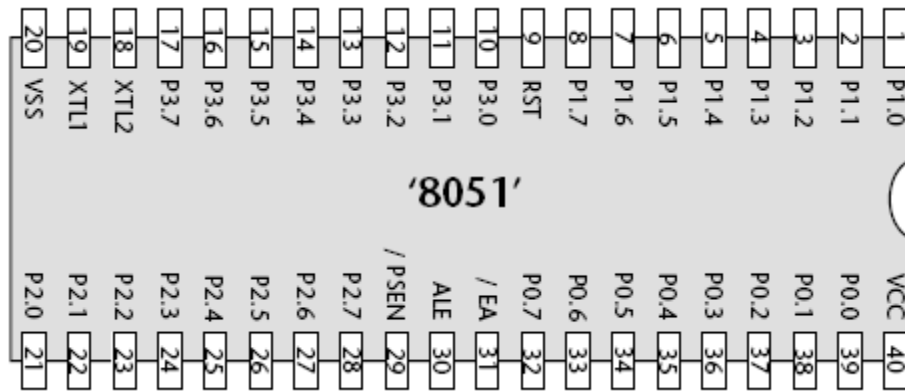
- Intel, Atmel, Philips / Signetics, Infineon, Dallas Semi/ Maxim

Why 8051?

- **Price** less than \$1.00 (US)
 - 1 MIPS performance
 - 256 **bytes** of on-chip RAM
- **Profile** - price, performance, available memory match needs
- 8051 **architecture** - by Intel
 - Philips, Infineon, Atmel and Dallas
 - Instruction Set / Code compatible
 - 60% microcontroller market - 50% of the 8-bit microcontroller market
 - Used in automotive systems to children's toys
- **low cost, huge range, easy availability and widespread use** of the 8051 family makes it
 - an **excellent platform for developing** embedded systems:
 - ideal platform for *learning* about embedded systems.

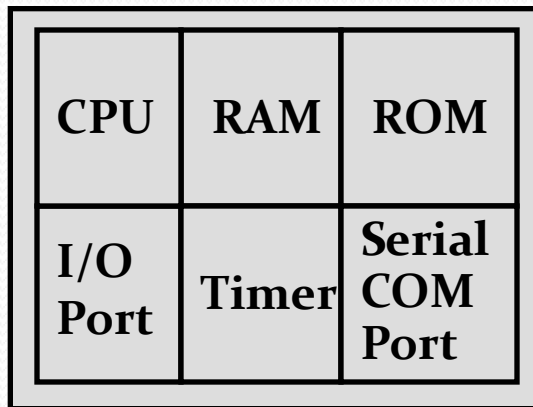
Intel MCS-51 family

- Standard 40-Pin DIP package
- Excellent device for building embedded systems
- Minimum # of external components to operate



8051 Basic Components

- 4K bytes internal **ROM**
- 128 bytes internal **RAM**
- Four 8-bit **I/O ports** (P₀ - P₃).
- Two 16-bit **timers**/counters
- One **serial** interface

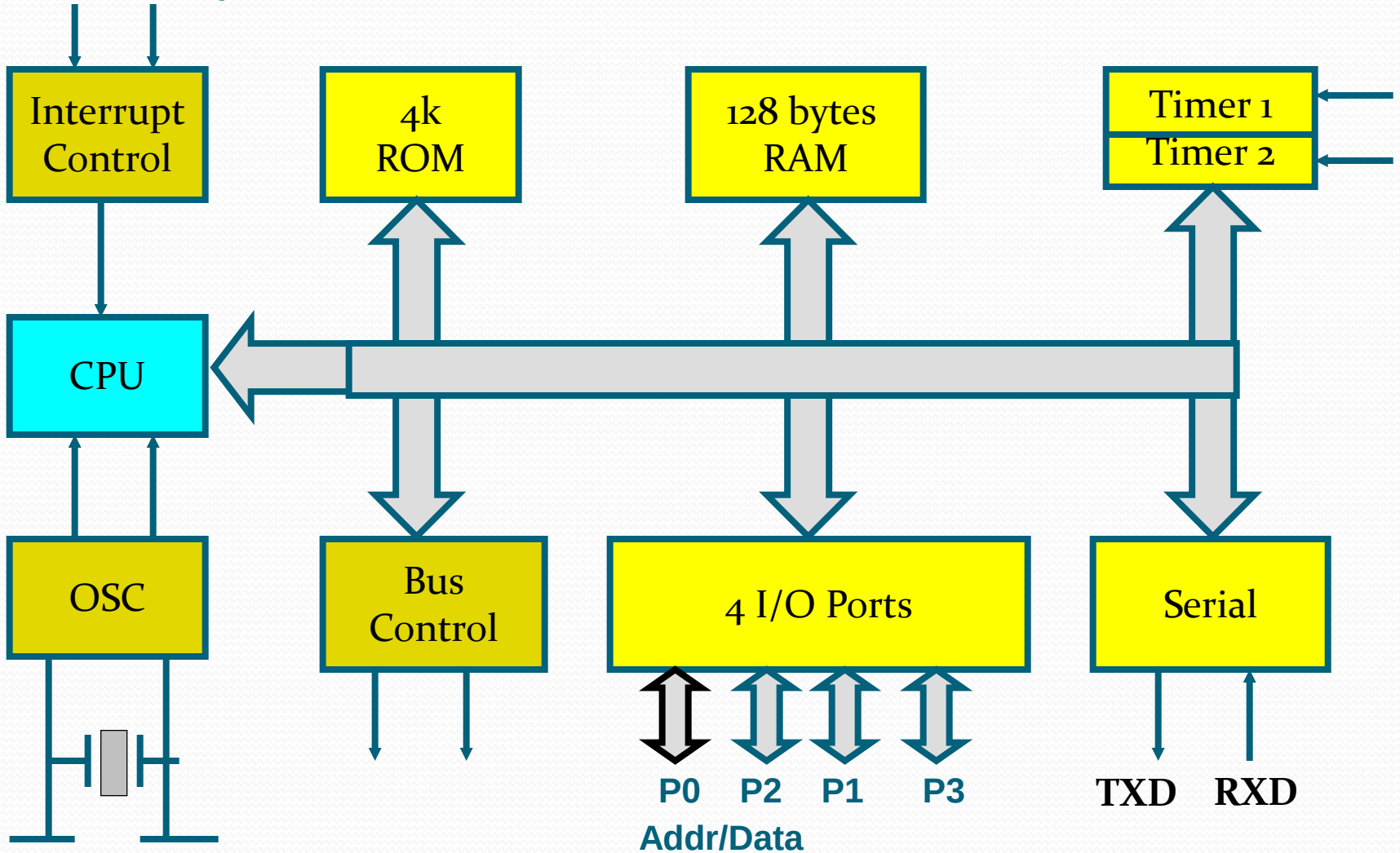


← A single chip
Microcontroller



Block Diagram

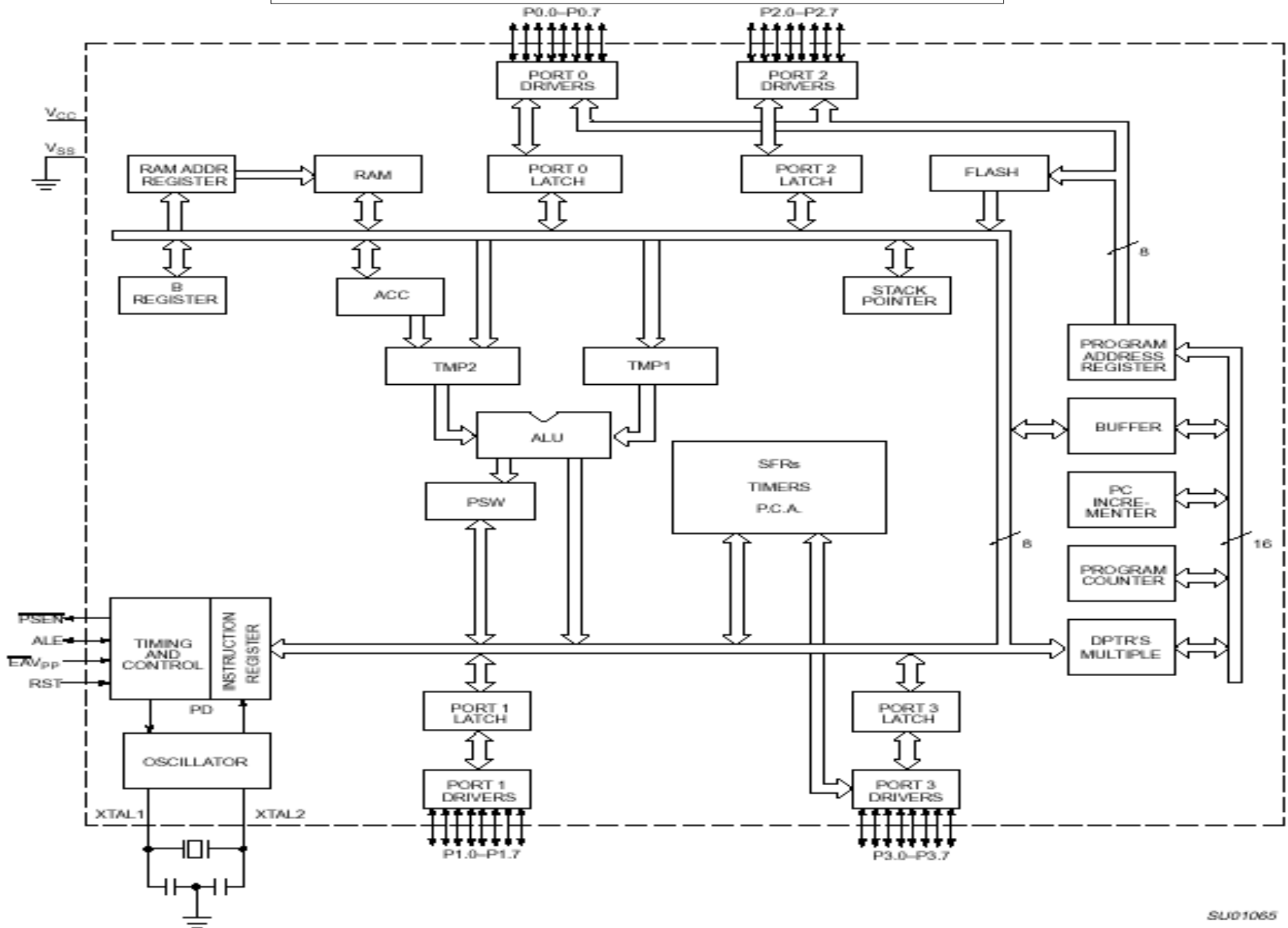
External Interrupts



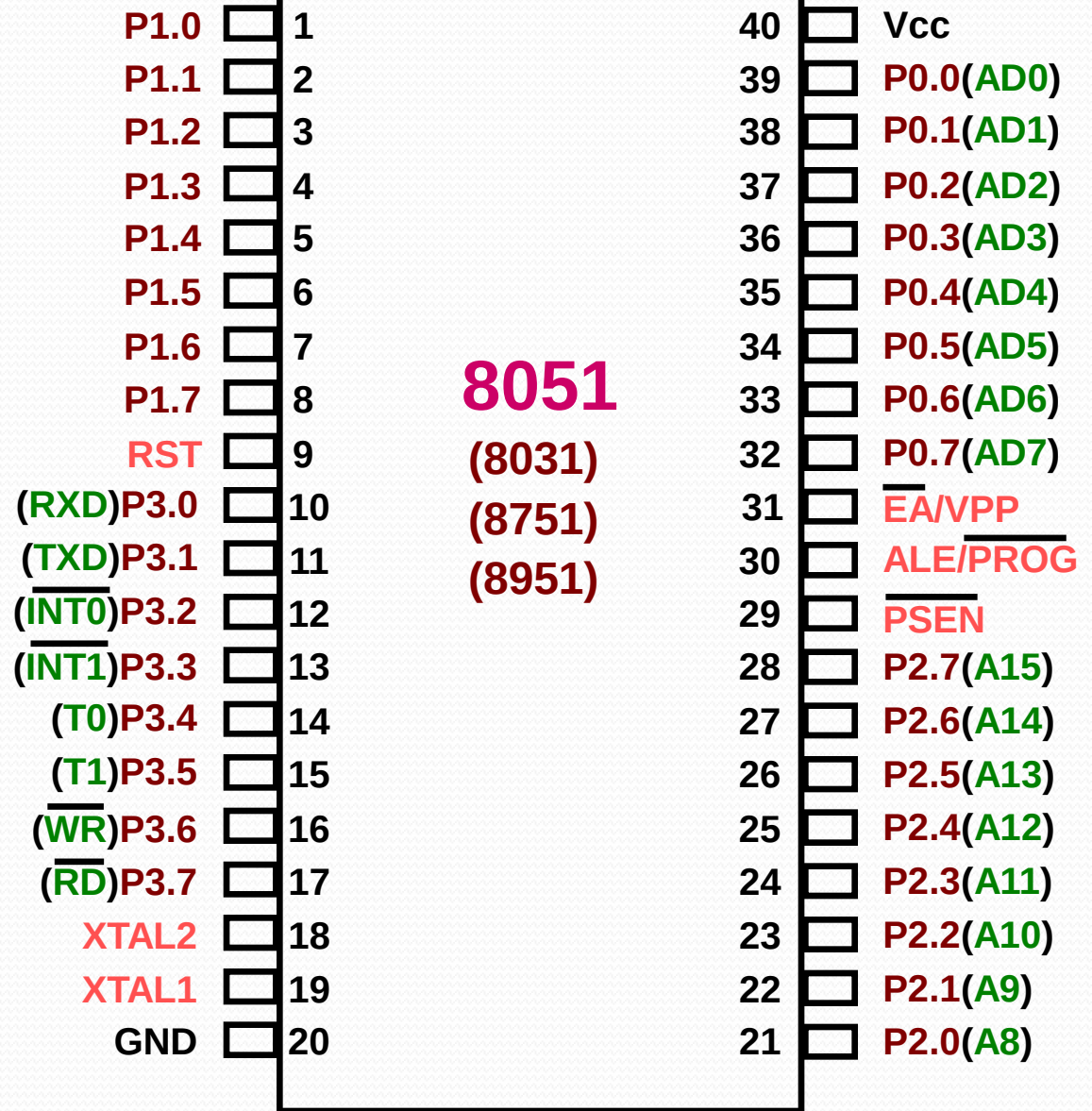
Other 8051 featur

- only 1 On chip oscillator (external crystal)
- 6 interrupt sources (2 external , 3 internal, Reset)
- 64K external code (program) memory(only read)PSEN
- 64K external data memory(can be read and write) by RD,WR
- Code memory is selectable by EA (internal or external)
- We may have External memory as data and code
- 21SFRs
 - 34 8-bit general purpose registers in total

8051 Internal Block Diagram



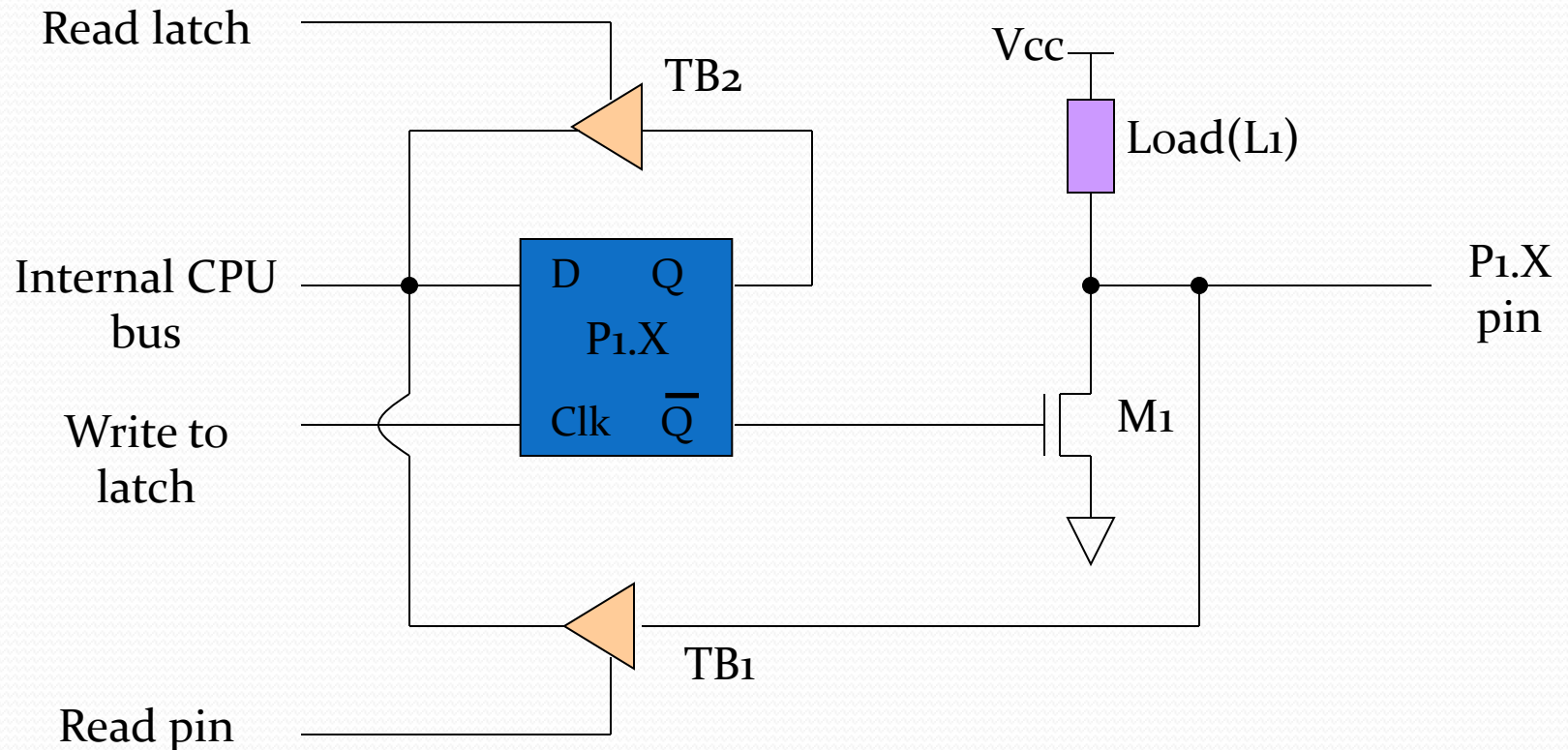
8051Foot Print



IMPORTANT PINS (IO Ports)

- One of the most useful features of the 8051 is that it contains four I/O ports (P0 - P3)
- Port 0 (pins 32-39) : P0 (P0.0~P0.7)
 - 8-bit R/W - General Purpose I/O
 - Or acts as a multiplexed low byte address and data bus for external memory design
- Port 1 (pins 1-8) : P1 (P1.0~P1.7)
 - Only 8-bit R/W - General Purpose I/O
- Port 2 (pins 21-28) : P2 (P2.0~P2.7)
 - 8-bit R/W - General Purpose I/O
 - Or high byte of the address bus for external memory design
- Port 3 (pins 10-17) : P3 (P3.0~P3.7)
 - General Purpose I/O
 - if not using any of the internal peripherals (timers) or external interrupts.
- Each port can be used as input or output (bi-direction)

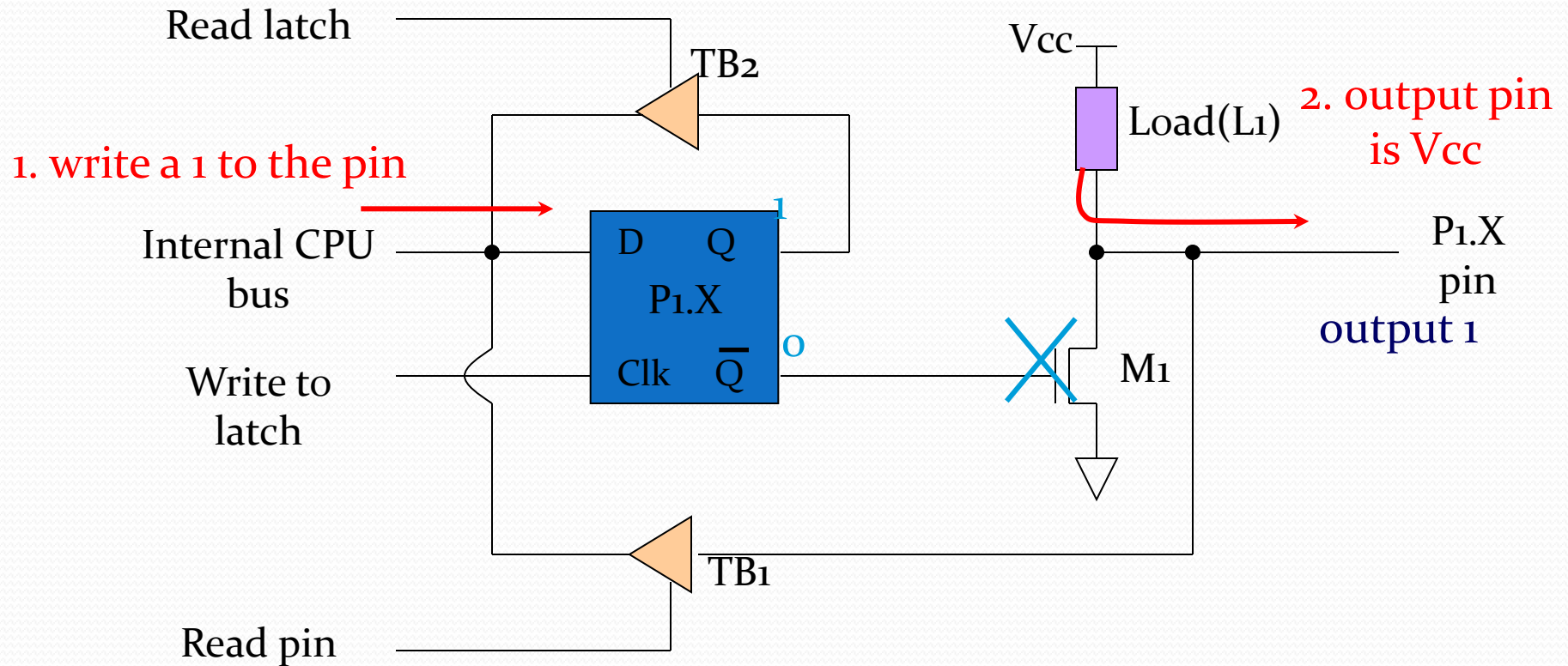
Hardware Structure of I/O Pin



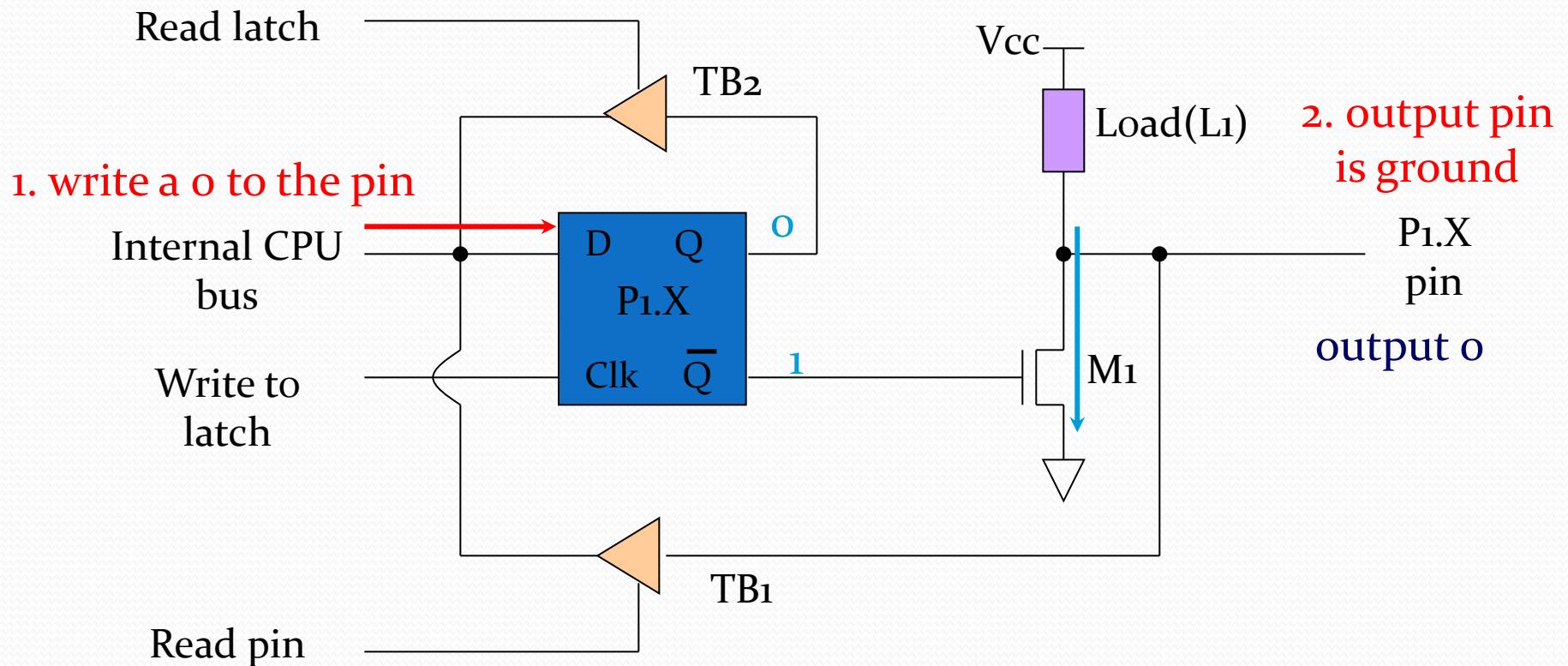
Hardware Structure of I/O Pin

- Each pin of I/O ports
 - Internally connected to CPU bus
 - A **D latch** store the value of this pin
 - Write to latch = 1 : write data into the D latch
 - 2 **Tri-state** buffer :
 - TB1: controlled by “Read pin”
 - Read pin = 1 : really read the data present at the pin
 - TB2: controlled by “Read latch”
 - Read latch = 1 : read value from internal latch
 - A **transistor** M₁ gate
 - Gate=0: open
 - Gate=1: close

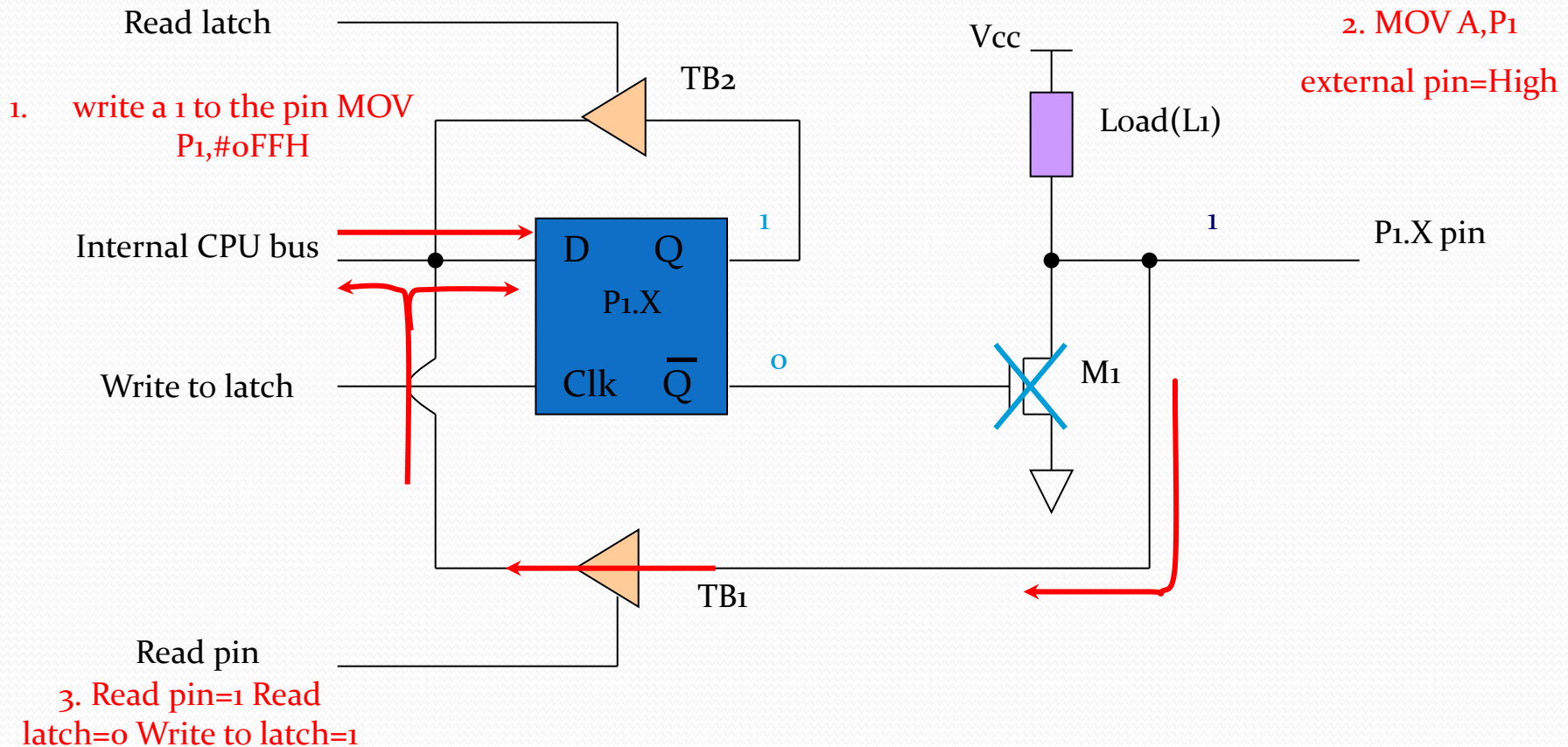
Writing "1" to Output Pin P1.X



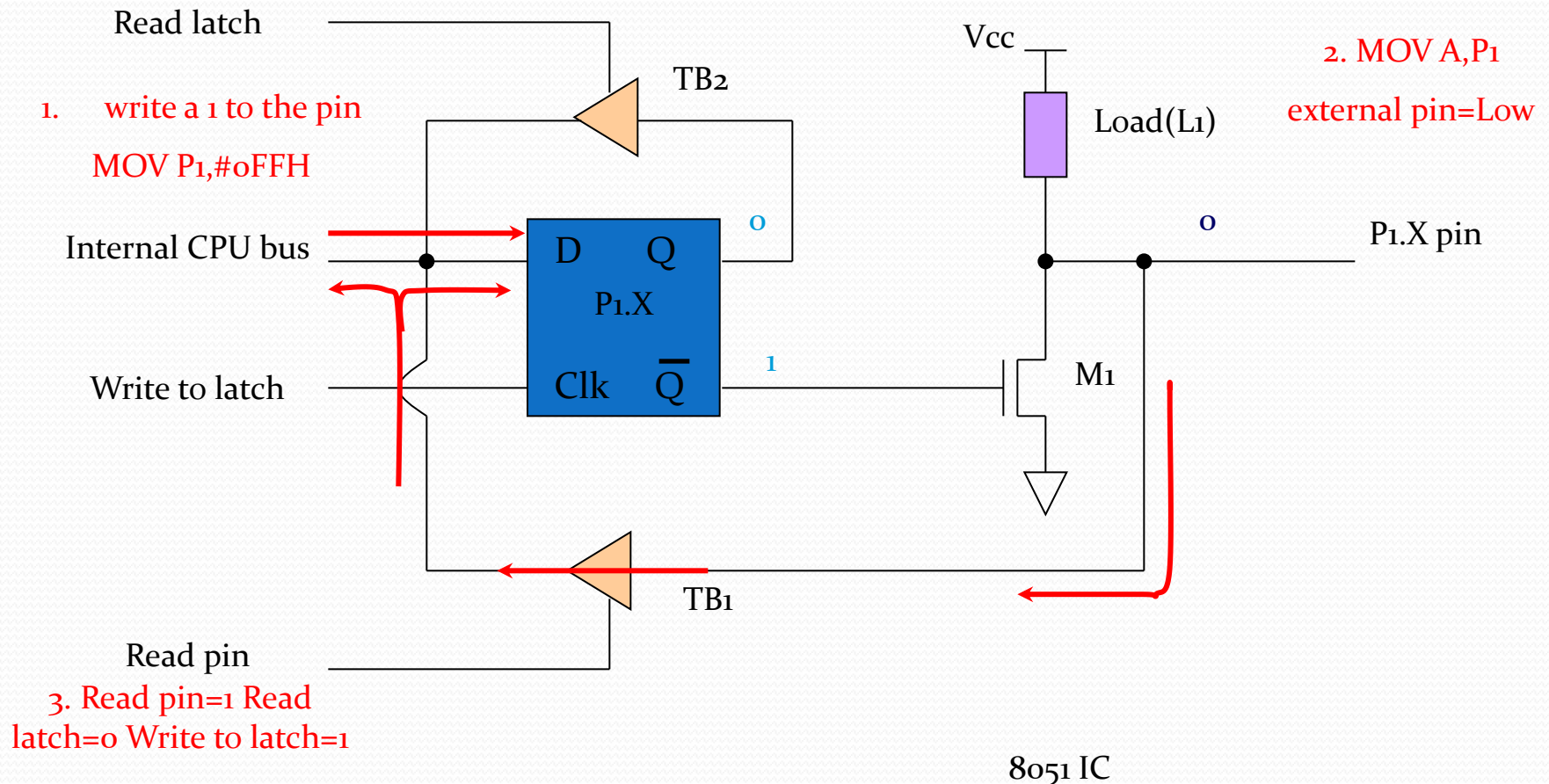
Writing "0" to Output Pin P1.X



Reading "High" at Input Pin



Reading "Low" at Input Pin



External Memory Interface

